higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENTATION TECHNOLOGY
NQF LEVEL 4

(6060474)

27 February 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL drawings must be neat and well balanced.
5. Clearly cross out ALL the work you do not want to be marked.
6. Start each section on a NEW page.
7. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Humidity is defined as the …

A mass per unit volume of a substance.
B force per unit area of a substance.
C amount of water vapour present in a sample.
D resistance to flow of a liquid.

1.1.2 A device that receives data from a measurement instrument and compares that data to a set point:

A Controller
B Interlock
C Recorder
D Positioner

1.1.3 A neutral solution has a pH value of …

A 0
B 7
C 14
D 1

1.1.4 A device used to shut off the supply of fuel upon a low-water signal in a boiler:

A Fan interlock
B Purge interlock
C Low-water interlock
D Loss-of-flame interlock

1.1.5 A diagram commonly used in engineering to indicate the general flow of plant processes and equipment is called:

A PID
B PFD
C PLC
D DCS

(5 × 2) (10)

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1.2 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–G) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	Until an unfired furnace has been air-cleansed no fuel may be admitted to the furnace	A	latching alarm
		B	feed-forward control
1.2.2	Difference between the measured variable of the controlled variable and the set point	C	offset
		D	deviation
1.2.3	Measure of effective acidity or alkalinity of solutions based on the concentration of hydrogen ions	E	purge interlock
		F	cascade control
1.2.4	Alarm where the relay cannot automatically reset	G	pH measurement
1.2.5	Control loop where the difference between the controlled variable and the set point need not exist for corrective action to occur		

(5 × 1) (5)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 A liquid with a pH of more than seven is acidic.

1.3.2 In a *closed-loop control system* there is no feedback signal fed to the controller which is compared with the set point of the process.

1.3.3 The first letter inside the circle on loop drawings that are used for instrument labelling, identifies the nature of the measured variable.

1.3.4 ORP measures the ratio of ions that oxidise to those that reduce other chemicals.

1.3.5 A load cell is a transducer which converts force into a measureable electrical output.

(5 × 1) (5)

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1.4 Give ONE term for each of the following descriptions. Write only the term next to the question number (1.4.1–1.4.5) in the ANSWER BOOK.

1.4.1 Cloudiness or haziness of a fluid caused by individual particles or suspended solids generally invisible to the naked eye

1.4.2 Device that is used to prevent a machine from damaging itself and from harming its operator by stopping the machine when it trips.

1.4.3 Electronic instrument used for measuring the acidity or alkalinity of a solution.

1.4.4 Digital computer used for automation of electromechanical processes such as control of machinery or factory assembling lines.

1.4.5 System comprising of a high-speed communications network, operations work centre, and microprocessor-based controllers.

(5 × 2) (10)
[30]

TOTAL SECTION A: 30

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SECTION B

QUESTION 2

2.1 Explain the operating principle of a hydrostatic level-measuring system. (9)

2.2 Name FOUR types of load cells. (4)

2.3 Name and explain the operation of THREE types of system configurations that can be used for open-tank level measurement using a differential pressure transmitter. (3 × 2) (6)

2.4 Name the THREE components that constitute a nuclear level-measurement system. (3)

2.5 Explain the operating principle of a nucleonic level detector. (8)
[30]

QUESTION 3

3.1 Explain the importance of in-line process analysers in a process-control environment. (9)

3.2 A dissolved-oxygen analyser is ONE of the in-line process analysers that is used in the process industry.

3.2.1 Discuss the basic working principle of a dissolved-oxygen analyser. (5)

3.2.2 Give TWO applications of a dissolved-oxygen analyser. (2 × 2) (4)

3.3 The moisture content of the atmosphere is of vital importance in many industrial processes.

3.3.1 Describe the construction of a dew-cell element that is used in a humidity analyser. (5)

3.3.2 Explain the operating principle of a dew cell analyser. (7)
[30]

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QUESTION 4

4.1 Name FOUR ways how turbidity in fluids can be measured. (4)

4.2 Draw a sketch with labels of an absorption turbidity analyser and explain its working principle. (13)

4.3 Name THREE basic purposes for using the pH and ORP levels of a liquid in a measuring process. (3)

4.4 Draw a simple sketch showing the major components of a typical pH analyser, and explain its working principle. (10)
[30]

QUESTION 5

5.1 Explain each of the following typical alarm configurations as used in control systems:

5.1.1 High-limit alarm

5.1.2 Low-limit alarm

(2 × 3) (6)

5.2 Give FIVE disadvantages of pneumatic signal-transfer elements. (5)

5.3 Explain the difference between an *open-loop control system* and a *closed-loop control system.* (7)

5.4 Draw signal-transfer diagrams for each of the following:

5.4.1 Process connection

5.4.2 Electrical signal

5.4.3 Pneumatic signal

(3 × 2) (6)

5.5 Explain the function of each of the following elements in a process-control system:

5.5.1 Control valve (2)

5.5.2 Positioner (4)
[30]

TOTAL SECTION B: 120
GRAND TOTAL: 150

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